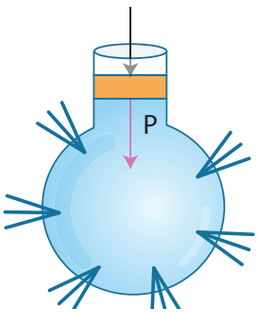
DO YOU KNOW?

In petrol bunks, the tyre pressure of vehicles is measured in a unit called psi. It stands for pascal per inch, an old system of unit for measuring pressure.

3.4 Pascal's Law

Pascal's principle is named after Blaise Pascal (1623-1662), a French mathematician and physicist. The law states that the external pressure applied on an incompressible liquid is transmitted uniformly throughout the liquid. Pascal's law can be demonstrated with the help of a glass vessel having holes all over its surface. Fill it with water. Push the piston. The water rushes out of the holes in the vessel with the same pressure. The force applied on the piston exerts pressure on water. This pressure is transmitted equally throughout the liquid in all directions (Fig. 3.8). This principle is applied in various machines used in our daily life.



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Figure 3.8 Demonstration of Pascal's Law

3.4.1 Hydraulic press

Pascal's law became the basis for one of the important machines ever developed, the hydraulic press. It consists of two cylinders of different cross-sectional areas as shown in Figure 3.9. They are fitted with pistons of cross-sectional areas "a" and 'A'. The object to be lifted is placed over the piston of large cross-sectional area A. The force F_1 is applied on the piston of small cross-sectional area 'a'. The pressure P produced by small piston is transmitted equally to large piston and a force F_2 acts on A which is much larger than F_1.

Pressure on piston of small area 'a' is given by,

$$P = \frac{F_1}{A_1} \quad (1)$$

Applying Pascal's law, the pressure on large piston of area A will be the same as that on small piston. Therefore, $P = \frac{F_2}{A_2}$ (2)

Comparing equations (1) and (2),we get

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$. or $F_2 = F_1 \times \frac{A_2}{A_1}$

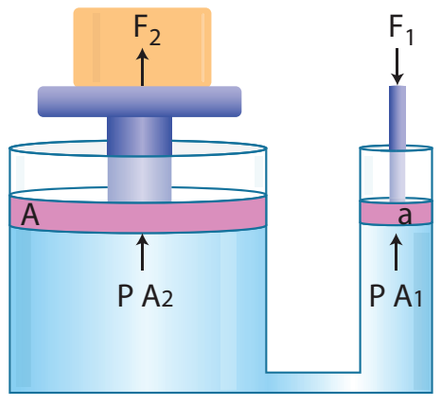


Figure 3.9 Hydraulic press

Since, the ratio $\frac{A_2}{A_1}$ is greater than 1, the force F_2 that acts on the larger piston is greater than the force F_1 acting on the smaller piston. Hydraulic systems working in this way are known as *force multipliers*.

Problem 4

A hydraulic system is used to lift a 2000 kg vehicle in an auto garage. If the vehicle sits on a piston of area 0.5 m^2, and a force is applied to a piston of area 0.03 m^2, what is the minimum force that must be applied to lift the vehicle?

Given: Area covered by the vehicle on the piston $A_1 = 0.5\ m^2$

Weight of the vehicle, $F_1 = 2000\ kg \times 9.8\ m\ s^{-2}$

Area on which force F_2 is applied, $A_2 = 0.03\ m^2$

Solution:

$P_1 = P_2$; $\frac{F_1}{A_1} = \frac{F_2}{A_2}$ and $F_2 = \frac{F_1}{A_1} A_2$;

$F_2 = (2000 \times 9.8) \frac{0.03}{0.5} = 1176\ N$

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